

## Assessing digital competence of K1-12 teachers in Kosovo: A study through the lens of DigCompEdu and TPACK

Krenare Pireva Nuci<sup>a</sup>, Fisnik Dalipi<sup>b,\*</sup>, Italo Masiello<sup>b</sup>, Ismail Celik<sup>c</sup>, Jari Laru<sup>c</sup>, Blerim Saqipi<sup>a</sup>, Arsim Susuri<sup>d</sup>, Ali Shariq Imran<sup>e</sup>

<sup>a</sup> Department of STEM, Faculty of Education, University of Prishtina, Kosovo

<sup>b</sup> Department of Informatics, Linnaeus University, Sweden

<sup>c</sup> Faculty of Education and Psychology, University of Oulu, Finland

<sup>d</sup> Faculty of Computer Science, University of Prizren, Kosovo

<sup>e</sup> Norwegian University of Science and Technology (NTNU), 2815, Gjøvik, Norway

### ARTICLE INFO

#### Keywords:

Education

Digital competences

DigCompEdu

Technology

TPACK

### ABSTRACT

Digital competence is a critical skill for teachers in the modern era, as technology reshapes the education ecosystem. The ability to integrate digital tools effectively into teaching practices not only enhances learning outcomes but also equips students with skills that are important for their daily lives. For teachers, fostering digital proficiency is fundamental to addressing the challenges of contemporary education environments, promoting inclusive learning, and addressing the evolving demands of the digital age. This article evaluates the digital competencies of K1-12 teachers in Kosovo using the DigCompEdu framework, providing information on six competency areas in the Digital Competence for Educators Framework (DigCompEdu). Using the SELFIE tool, data were collected from 441 in-service teachers across Kosovo and analyzed through normality and reliability tests. The Kolmogorov-Smirnov (KS) and Shapiro-Wilk (SW) resulted in significant deviations from normality; therefore, the analysis proceeded with non-parametric methods, primarily using the Kruskal-Wallis test for group comparisons. The reliability resulted in strong internal consistency with a Cronbach's Alpha value of 0.901. The results reveal varied proficiency levels, with the highest scores in the "Digital Resources" domain and the lowest in "Teaching and Learning". Age and experience with digital tools significantly influenced certain competencies, highlighting disparities between demographic groups. By aligning DigCompEdu with the TPACK framework, this research underscores the need to bridge the "what" (skills) with the "how" (application) in developing digital competence for effective integration by teachers.

### 1. Introduction

The advancement of technology and the integration of cutting-edge technology in schools mark a significant milestone, shifting traditional teaching and learning toward technology integration in education. This shift underscores the need for the digital transformation of schools, a process also emphasized by Masiello et al. for the context of Sweden [1]. The effectiveness of digital transformation in education depends on supportive policy frameworks, professional development initiatives, and inclusive access to digital resources. The international institutions such as the UNESCO and OECD emphasize the importance of creating future-ready teachers and comprehensive system-level policies to

maximize the benefits of digital technologies [2]. Whereas, the vision for digital education for European schools emphasizes the need to develop digital competence in all learners, enabling the confident, critical, responsible, and creative use of digital technologies for learning and work [3]. That is why, in 2018, the European Commission adopted a Digital Education Action Plan to support technology use and digital competence development in education<sup>1</sup> as digital competence is one of the key competencies in a modern society.

Lakkala et al. [4] refer to digital competence as the ability to use technology to manipulate information literacy, digital literacy, media literacy, and Internet literacy for their own purposes. The research in [5] listed seven core skills that build up digital competencies: technical,

\* Corresponding author at: Department of Informatics, Linnaeus University, P G Vejdes väg, 351 95, Växjö, Sweden.

E-mail address: [fisnik.dalipi@lnu.se](mailto:fisnik.dalipi@lnu.se) (F. Dalipi).

<sup>1</sup> Digital Education Action Plan; URL: <https://education.ec.europa.eu/focus-topics/digital-education/action-plan>

information management, communication, collaboration, creativity, critical thinking, and problem-solving. In line with [5], Pedaste et al. [6] claim that digital competence extends beyond mere technological literacy, encompassing the ability to critically engage with digital content, solve problems, collaborate, and innovate using digital tools across educational and professional domains.

In the digital education report, “Teachers” is one of four key entities, with subcategories “Teacher-specific digital competencies in initial teacher education (ITE)” and “Alternative pathways and retraining for specialized informatics teachers” to strengthen the digital education ecosystem [7]. This is why Redecker (2017), proposes to integrate digital competencies into teacher education, and professional teaching training programs to prepare teachers to act as role models and effectively use digital technologies in teaching and learning [8].

In this regard, as discussed by [9,10], the European Digital Competence Framework (DigComp) and DigCompEdu for educators framework with respect to digital competence involves areas related to problem-solving, collaboration, information literacy, and digital content creation - to name a few. This comprehensive approach is essential to addressing the demands of an increasingly digital world. To promote the use of technology in education, there is a need to develop first digital competences of teachers, and as the EU aims to all European citizens.<sup>2</sup>

Another theoretical model approach alongside with DigCompEdu, the technological pedagogical content knowledge (TPACK), serves as a model for understanding how teachers integrate technology with pedagogy by combining three core elements: Technological Knowledge (TK), Pedagogical Knowledge (PK), and Content Knowledge (CK) [11]. This framework helps school teachers one step further by effectively selecting and applying digital tools, ensuring a balanced approach to subject-specific teaching.

Both models highlight comprehensive digital competence as an essential skill for integrating digital technologies into education, and this is why, alongside the development of the infrastructure, the adaptability of teachers to new digital competencies has become a critical priority.

Shifting the discussion to the integration of digital literacy into the K1-12 curricula, many countries have already integrated digital literacy into their curricula, acknowledging it as a core component of 21st-century education. The digital competence is integrated into K1-12 education systems through (i) cross-curricular, (ii) separate subject, or (iii) integrated approaches [7]. Countries like Turkey, Denmark, Greece, Spain, Italy, Estonia, and, recently, Albania introduce it from the first grade as a compulsory separate subject, whereas Croatia, Hungary, and Romania do from the 5th grade. Nordic countries, including Sweden and Finland, revised K1-12 curriculum highlights programming and digital competence as interdisciplinary elements across subjects like mathematics, social sciences, and technology [12]. Kosovo introduces digital competence as a compulsory separate subject from the 6th grade. However, assessing whether the curriculum is up to date and includes cutting-edge technologies is beyond the scope of this paper.

Given these developments, whether driven by technological advancements or the integration of digital competencies into the curriculum, it is crucial to evaluate whether current school teachers have the necessary digital skills, require professional development, or need a specialized profile in digital competence. However, significant challenges persist. Studies highlight ongoing issues such as unequal access to technology, limited digital proficiency among teachers, and a lack of tailored training programs that address diverse learning contexts. According to [13], addressing these challenges requires a coordinated approach that involves policymakers, teachers, and other stakeholders to create inclusive adaptive digital ecosystems and ensure sustained improvement in digital competence.

In light of this, the aim of this research paper is to investigate the digital competencies of Kosovo pre-university K1-12 teachers within the framework of DigCompEdu, providing a detailed overview of their current digital skills, access to resources, and willingness to promote their students’ digital skills.

Furthermore, the findings of this study are analyzed in the context of education systems in EU countries where established digital competence standards provide a valuable benchmark for comparison. By examining these factors, this research aims to provide insight into the potential for improvements for digital competence in the educational landscape of Kosovo and in line with OECD (2023) to contribute to broader discussions on best practices for improving the school teacher digital competencies through a nationally coordinated approach.

Despite the growing body of research on teachers’ digital competence, several important limitations remain. First, much of the existing literature focuses on well-established educational systems, particularly within EU and OECD contexts, while less attention has been given to countries such as Kosovo, where digital education policies and practices are still developing [14]. As a result, there is limited empirical evidence on how teachers in such contexts perceive and act on their digital competence. Second, although frameworks such as DigCompEdu provide comprehensive descriptions of teachers’ digital competence, prior studies have primarily emphasized technical and resource-related dimensions, often reporting higher competence in areas such as digital resource management, while consistently identifying weaker performance in pedagogical integration (e.g., teaching, learning, and learner empowerment) (e.g., [15]). However, these studies rarely go beyond descriptive reporting to explain why this imbalance occurs or how it relates to broader models of teacher knowledge. Third, existing research tends to rely heavily on self-report instruments, such as SELFIE, without sufficiently connecting the results to theoretical models that explain how technology, pedagogy, and content interact in practice [16]. This creates a gap between measuring digital competence (what teachers report they can do) and understanding how this competence translates into meaningful instructional practice. Finally, while DigCompEdu is widely used for assessment, it does not explicitly model the integration of technological, pedagogical, and content knowledge, which is central to frameworks such as TPACK [17]. As a result, there is a need to complement DigCompEdu-based findings with a framework that can better capture the pedagogical meaning of digital competence.

The remainder of this paper is structured as follows: Section 2 provides a comprehensive synthesis of the theoretical and empirical background, starting with the evolution of digital literacy and concluding with the specific aim and research questions that guided this study. Section 3 details the research design, including the quantitative methodology and the application of the SELFIE for teachers framework in the Kosovo context. Section 4 presents the results of the statistical analysis, whereas Section 5 discusses these findings in the context of previous research and interprets the results through the lens of the TPACK framework. Finally, Section 6 provides concluding remarks, summarizes the study’s contributions, and suggests avenues for future work.

## 2. Background and related work

### 2.1. The notion of digital literacy and digital competence

Conventional methods for fostering digital competencies in teacher education emphasize ‘digital literacy’ [18]. The word initially appeared around 1997, when Paul Gilster [19] presented it in his publication as: . . . a set of skills to access the internet, find, manage, and edit digital information; join in communications; and otherwise engage with an online information and communication network. Digital literacy is the ability to properly use and evaluate digital resources, tools, and services, and apply them to lifelong learning processes.

As new technologies and their applications have emerged due to the widespread availability of the internet and the development of personal

<sup>2</sup> DigCompEdu; URL : <https://joint-research-centre.ec.europa.eu/dig-compedu/en>

mobile digital devices, the term has become more contentious. Terminology such as 'information literacy' [20], 'computer literacy' [21], 'internet literacy' [22], 'media literacy' [23], and more recently, 'multi-modal literacy' [24] has been linked to the proficient utilization of digital resources in education and has been advocated as integral elements of a comprehensive perspective on digital literacy. Menon et al. [25] note that establishing a definitive definition of digital literacy is difficult, owing to the continuously changing technical, cultural, and sociological contexts that redefine the use of digital technology in both personal and professional spheres.

In teacher education, cultivating digitally literate students has typically involved prioritizing technical skills for utilizing digital tools and systems considered suitable for educational contexts, as well as determining their application within specific learning units [25,26]. This method posits that it equips pre-service teachers with a set of basic competencies they can transfer to their future classrooms. These methodologies have been critiqued by [25,27,28] for their limited emphasis on skills, absence of authenticity, disregard for diverse socio-cultural contexts in technology utilization, and their ineffectual, simplistic design. Whereas in [29,5] are recognized shortcomings in their excessively technical methodology, which overlooks broader factors such as ethics, digital citizenship, health, welfare, safety, and social/collaborative aspects.

In considering the notion of digital competence, Janssen et al. [30] define digital competency as a set of skills involving more than knowing how to use devices and applications; it also includes the ability to interact with information and communication technology (ICT) and information literacy skills. This conceptualization recognizes the significance of technical knowledge and abilities while adopting a larger socio-cultural perspective that emphasizes the need for understanding the wider implications and impacts of digital technology on people and society.

## 2.2. Frameworks for digital competencies of teachers

Developing digital competencies in teachers is essential for effective teaching in today's technology-driven educational landscape. In recent years, there has been a global push to develop frameworks that support and assess digital competencies in education, particularly for teachers. This trend reflects the growing recognition of technology's crucial role in modern pedagogy. On a global scale, UNICEF developed the Digital Competency Framework for teachers,<sup>3</sup> focusing on empowering teachers with digital and innovation competencies. Similarly, in Europe, the European Training Foundation created the ETF READY Model in 2022.<sup>4</sup> This model serves as a reference for teachers' activities and development in the 21st century.

Many developed countries have also implemented national frameworks. As highlighted in [31], Spain adapted the DigComp 2.1 and DigCompEdu frameworks to create the Common Digital Competence Framework for Teachers (CDCFT). As discussed in [32,33], Norway introduced the Professional Digital Competence Framework for Teachers in 2017 to enhance teacher education and professional development. It equips teachers to integrate digital tools into education by embedding digital competence in teacher training and professional development, focusing on seven key areas such as subject integration, digital ethics, communication, and leadership in digital learning, among others. The framework aligns with Norwegian national policies and EU initiatives like DigCompEdu, reinforcing a commitment to innovation and digital literacy. The United Kingdom established the Digital Competence

Framework and the Digital Teaching Professional Framework to support both teachers and citizens in developing digital skills.<sup>5</sup> UNESCO's recent framework, "Artificial Intelligence and Digital Transformation Competencies for Civil Servants,"<sup>6</sup> promotes the use of AI technologies in education. This initiative emphasizes the importance of ensuring AI integration is legally, ethically, and technically sound. For a more detailed EU initiative, frameworks designed to increase digital competencies initially for a wider group and continue for the teacher and learners group evolved year by year; the final models are listed in Table 1. The initial Digital Competence Framework for Citizens (DigComp), comprised of 21 competences organized in five competence areas, emerged from the Europe 2020 Strategy by emphasizing the need for the digital competence to benefit from digital opportunities, such as employability, quality education, social inclusion and poverty reduction [34].

In 2017, the Digital Competence Framework for Educators (DigCompEdu) emerged as a scientifically sound framework designed only for teachers. The framework is comprised of 22 competencies organized in six areas and described through 6 proficiency levels (A1 Awareness, A2 Exploration, B1 Integration, B2 Expertise, C1 Leadership, and C2 Innovation). Table 2 lists the areas and the competencies. In addition, it presents the link between the profiles, areas, and competencies. It is important to emphasize that DigCompEdu, designed and presented by [8], serves as a foundational framework for all upcoming national frameworks, such as:

**Table 1**  
EU initiatives for digital competences.

| Framework                                                 | Description                                                                                                                                                             | Provider/<br>Year | Target                                                   |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------|
| DigCompEdu 3.0                                            | A standardized framework for identifying and measuring digital skills across society, work, and education.                                                              | EU/2026           | Citizens                                                 |
| DigComp 2.2                                               | A framework that supported the development of digital competences of individuals personally and professionally.                                                         | EU/2022           | Policy makers, teachers/trainers, labour market partners |
| Common Framework of Reference for Intercultural (CFRIDIL) | A framework for standardising digital skills by promoting transparency and recognition for the evaluation of what an actor should in transnational digital environments | EU/2019           | teachers/trainers, citizens, NGOs and researchers        |
| DigCompEdu                                                | A framework that supported the development of digital competences to teachers.                                                                                          | EU/2017           | Teachers                                                 |
| DigCompEdu 2.1                                            | A framework that supported the development of individuals personally and professionally.                                                                                | EU/2017           | Citizens                                                 |
| DigCompEdu 2.0                                            | The Conceptual Reference Model                                                                                                                                          | EU/2016           | Citizens                                                 |

<sup>3</sup> UNICEF Framework; URL: <https://www.unicef.org/eca/media/24526/file/Educators>

<sup>4</sup> ERF Ready Model, <https://openspace.etf.europa.eu/content/ready-model>

<sup>5</sup> DCF; <https://connect.unevoc.unesco.org/home/Digital+Competence+Frameworks/lang=en/id=17>

<sup>6</sup> <https://unesdoc.unesco.org/ark:/48223/pf0000383325>

**Table 2**  
DigCompEdu framework composition.

| Profiles                           | Area                                         | Competences                                            |
|------------------------------------|----------------------------------------------|--------------------------------------------------------|
| Educators professional competences | 1. Professional Engagement                   | 1.1 Organisational communication                       |
|                                    |                                              | 1.2 Professional Collaboration                         |
|                                    |                                              | 1.3 Reflective Practice                                |
|                                    |                                              | 1.4 Digital Continuous Professional Development        |
| Educators pedagogical competences  | 2. Digital Resources                         | 2.1 Selecting digital resources                        |
|                                    |                                              | 2.2 Creating and modifying digital content             |
|                                    |                                              | 2.3 Managing, protecting and sharing digital resources |
|                                    | 3. Teaching and Learning                     | 3.1 Teaching                                           |
|                                    |                                              | 3.2 Guidance                                           |
|                                    |                                              | 3.3 Collaborative learning                             |
|                                    |                                              | 3.4 Self-regulated learning                            |
|                                    | 4. Assessment                                | 4.1 Assessment strategies                              |
|                                    |                                              | 4.2 Analysing evidence                                 |
|                                    |                                              | 4.3 Feedback and Planning                              |
| Learners' competences              | 5. Empowering Learners                       | 5.1 Accessibility and inclusion                        |
|                                    |                                              | 5.2 Differentiation and personalisation                |
|                                    |                                              | 5.3 Actively engaging learners                         |
|                                    | 6. Facilitating Learners' Digital Competence | 6.1 Information and media literacy                     |
|                                    |                                              | 6.2 Digital communication and collaboration            |
|                                    |                                              | 6.3 Digital content creation                           |
|                                    |                                              | 6.4 Responsible use                                    |
|                                    |                                              | 6.5 Problem Solving                                    |
|                                    |                                              |                                                        |

- the Spanish Common Framework for the Digital Competence of Teachers (CDCFT) ,<sup>7</sup>
- the Austrian digi.kompP model [35],
- the Norway Professional Digital Competence Framework for Teachers, to name a few as stated by [33].

To evaluate the digital competency of primary and secondary education teachers, instruments were designed based on the theory of knowledge, ability, and attitudes of the digital competency frameworks. Such an instrument that is widely used today is the SELFIE for Teachers questionnaire (SELFIE stands for Self-reflection on Effective Learning by Fostering Innovation through Educational Technology), which is an online tool designed by a team of experts from schools, education ministries and European research institutes, and funded by the European Commission.<sup>8</sup>

According to the website, SELFIE for Teachers has been used by over 100 000 users.<sup>9</sup> SELFIE for Teachers, is an individual self-assessment tool comprised of questions that cover the six areas of the DigCompEdu framework. A further iteration of the SELFIE is the 'CheckIn HE' tool,<sup>10</sup> which focuses on higher education educators. In addition to the six areas of DigCompEdu HE incorporates a seventh area, Open Education, which comprises of (i) open educational resources, (ii) open educational practices, and (iii) open science. The Open Education area is based on the Open Edu Framework (JRC 2016, 2019), through which researchers and HE teachers self-assess their use of digital tools to improve their teaching and research work.

All of these frameworks share a common goal: to define the qualities of a digitally competent teacher, acknowledging that technology has become an integral part of everyday work routines. As a result, many countries have assessed the digital competencies of their teachers, recognizing the direct impact on the quality of education through digital technology. The findings of these assessments can guide the development of targeted training programs, enabling teachers to gain confidence in the use of digital technologies - not just in collaboration with their colleagues, but especially with their students.

### 2.3. Assessing teachers' digital competencies by the frameworks

DigCompEdu was adapted in [36] to conceptualize teachers' digital literacy as a basic skill and the ability to use technology that strongly impacts the effectiveness and quality of teaching in China. In addition, it assessed six areas of competence to use digital technologies for educational improvement and innovation.

The Finnish Framework for Digital Competence developed under the Right to Learn program, it provides guides, videos, and models to help teachers integrate digital skills effectively. It ensures equal access to digital skills for learning, work, and social participation and it promotes teacher professional development for teaching effectively through ICT.<sup>11</sup>

According to [37], Portuguese K1-12 teachers who answered the SELFIE instrument showed overall a moderate level of digital competence, with the lowest scores observed in the areas of Assessment, Empowering Learners, and Facilitating the development of Learners' Digital Competence. SELFIE was also collected in Sweden in a pilot study by [38]. The researchers determined that participating teachers showed a realistic self-estimation of the level of digital competence, the majority showing a lower self-estimation, while only a few overestimating their competence compared to that of the SELFIE instrument itself. The same researchers also used observations and log books to confirm the results.

Kosovo is notably absent from the country lists of online self-assessment platforms such as SELFIE, Check In HE, and similar tools. This absence motivated the present study, which aims to provide scientific insights into the assessment of teacher digital competence in Kosovo.

### 2.4. Integration of technology into teaching and learning

Researchers in [39,40] with respect to digital competence have used existing frameworks to examine and enhance teachers' integration of technology into classroom activities. Among these, two conceptual frameworks stand out for their widespread use and strong empirical foundation: SAMR (Substitution, Augmentation, Modification, Redefinition) and TPACK.

The SAMR framework is a more hierarchical use of technology in education: from Substitution, replacing conventional resources with digital ones, to redefinition, curriculum, pedagogy, and practice reconceptualized with digital tools/technology. In SAMR, substitution and augmentation relate to enhancement, whereas modification and redefinition lead to transformation [41]. Although the SAMR framework offers descriptive "aim points" for pre-service and practicing teachers to improve their practice, it lacks specific examples of practices that could be used for each stage, methods for moving from one stage to another, and it doesn't take into consideration the necessary and supportive changes in pedagogy, technology, and learning design.

The TPACK model proposed by [42] provides an alternative, more comprehensive framework that attempts to solve SAMR's limitations. In contrast to SAMR, it does not depict a hierarchy or tiered development. TPACK harmonizes all these components into a single framework that combines strong conceptual knowledge (CK) with an awareness of how technology can be used to improve education (TCK) via pedagogies that build on students' existing knowledge and address their individual learning gaps.

Aligning TPACK with DigCompEdu is essential to create a comprehensive framework that addresses both the skills needed by K1-12 teachers and their application in the classroom [43]. This alignment promotes a holistic, practical, and transformative approach to technology integration in education, ensuring that digital competence frameworks are not only theoretical but also actionable and context-sensitive.

<sup>7</sup> [https://intef.es/wp-content/uploads/2023/04/English-SFDCT\\_2022.pdf](https://intef.es/wp-content/uploads/2023/04/English-SFDCT_2022.pdf)

<sup>8</sup> <https://education.ec.europa.eu/selfie/about-selfie>

<sup>9</sup> <https://education.ec.europa.eu/selfie-for-teachers>

<sup>10</sup> <https://ec.europa.eu/eusurvey/runner/CheckInHEv2021EN>

<sup>11</sup> [https://eperusteet.opintopolku.fi/#/en/digiosaaminen/8706410/osaamisen\\_kokonaisuus/8706431](https://eperusteet.opintopolku.fi/#/en/digiosaaminen/8706410/osaamisen_kokonaisuus/8706431)



This synergy supports more effective teacher training, policy-making, and research, driving innovation and improvement in education.

Aligning the two frameworks bridges the “what” (skills) with the “how” (application), creating a comprehensive approach to evaluating and developing teachers’ digital competence. DigCompEdu focuses on individual competencies, such as selecting digital tools or designing inclusive activities, but does not emphasize a systematic view of how these competencies interact with pedagogy and content knowledge [44]. TPACK fills this gap by emphasizing the intersection of technological, pedagogical, and content knowledge, enabling a deeper understanding of how K1-12 teachers can integrate technology effectively in subject-specific contexts [45]. By aligning these frameworks, the authors of the framework assess not only the possession of digital skills but also their contextualized and meaningful use in the classroom [46].

The alignment of TPACK with DigCompEdu for the interpretation of the findings of this research paper ensure that the frameworks can be tailored to meet the unique challenges of teaching different subjects and addressing the diverse needs of learners.

2.5. Aim and research questions

While the preceding literature establishes frameworks for digital competence, there remains limited empirical evidence in developing contexts like Kosovo and the notable gap between reporting technical skills and understanding their pedagogical integration through models like TPACK. To address these, the aim of this research paper is to explore the digital competencies of K1-12 teachers in Kosovo, using SELFIE as a self-reflection tool developed by the EU Science Hub [37]. Notably, online self-assessment tools such as SELFIE are missing Kosovo from their list of countries. To reach our aim, we established three research questions:

- **RQ 1:** What are the ICT competence levels of Kosovo’s K1-12 teachers based on the DigCompEdu framework?
- **RQ 2:** How do digital competencies, as defined by the DigCompEdu framework, vary between different age categories, levels of technology experience, and major cities in Kosovo?
- **RQ3:** How does the DigCompEdu framework compare to TPACK in integrating emerging technologies to prepare K1-12 teachers for future innovations?

Research question 1, addresses the lack of empirical evidence in underrepresented contexts such as Kosovo by examining teachers’ digital competence across the DigCompEdu areas. Research question 2, focuses on the limited understanding of how contextual and demographic factors shape digital competence by exploring differences related to age, experience with digital technology, and municipality. Research question 3, addresses the theoretical gap between measuring digital competence and explaining its pedagogical integration by interpreting the findings through the lens of TPACK.

3. Research design

This study adopts a quantitative research approach to assess the digital competencies of K1-12 teachers using the SELFIE for Teachers framework. The instrument as used in [8] comprises 22 competences organized into six areas as shown in Table 2. For each area, the number of items is as follows: Professional Engagement (4 items), Digital Resources (3 items), Teaching and Learning (4 items), Assessment (3 items), Empowering Learners (3 items), and Facilitating Learners’ Digital Competence (5 items). Each item within a specific area is measured using a Likert scale, and responses are subsequently aggregated and averaged to produce area-level mean scores. The resulting mean scores were further interpreted using the DigCompEdu proficiency scale as shown in Table 3, where predefined score ranges correspond to levels from A1 (Newcomer) to C2 (Pioneer). Missing data were minimal. Only

**Table 3**  
DigCompEdu proficiency levels.

| Competence categories | Mean score |
|-----------------------|------------|
| A1 (Newcomer)         | 1.0 – 1.8  |
| A2 (Explorer)         | 1.9 – 2.6  |
| B1 (Integrator)       | 2.7 – 3.4  |
| B2 (Expert)           | 3.5 – 4.2  |
| C1 (Leader)           | 4.3 – 4.7  |
| C2 (Pioneer)          | 4.8 – 5    |

two participants (0.45% of the sample) contained missing responses. Cases with more than 20% missing responses across the SELFIE for Teachers items were designated for exclusion; however, no participant exceeded this threshold. Therefore, all cases were retained for analysis, and the few missing values were addressed using item-level mean imputation. To ensure content validity, the instrument was translated from English into Albanian and subsequently back-translated into English, and it was reviewed and validated by three levels of experts to ensure content accuracy and semantic equivalence.

3.1. Data collection

The online survey has been disseminated by the Faculty of Education established professional network with school directors across municipalities in Kosovo using a snowball sampling method. Through this network, initial contact was made with directors in municipalities with stronger institutional collaboration in a total of 26 municipalities, who subsequently disseminated the survey internally to their teaching staff. In this context, the use of snowball sampling refers to this structured referral process. The sample consists of 441 teachers out of an estimated population of 20,787 pre-university teachers in Kosovo, representing approximately 2.1% of the total population. In addition, participants were drawn from 26 out of 38 municipalities, covering around 68% of all municipalities in Kosovo. From a sampling perspective, the sample size achieved is considered adequate for survey-based research. Contemporary methodological guidelines indicate that, for large populations, a sample size of approximately 385 respondents is sufficient to achieve a 95% confidence level with a 5% margin of error, with larger samples improving precision and stability [47,48]. Therefore, the sample of 441 exceeds commonly recommended thresholds and supports the statistical reliability of the findings. This indicates that the study achieves broad geographic coverage and a meaningful sample size, allowing for a reasonably diverse representation of teachers across regions.

Ethical considerations were addressed by ensuring informed consent, confidentiality, and anonymity of participants, in accordance with institutional ethics standards. Information about the study, including its purpose and voluntary nature, was provided within the survey, and participants indicated their consent prior to participation.

The survey results classify digital competencies into six proficiency levels, as presented in Table 3. These levels range from A1, denoting K1-12 teachers at the “Newcomer” stage of digital competence, to C2, representing “Pioneers” who demonstrate advanced proficiency in digital skills. This categorization provides a framework for assessing and interpreting the varying levels of digital competence among K1-12 teachers. What is important to notice is that the two highest levels denote a shift from self-realization of the teacher to a focus on students rather than the self, like for example, passing on their knowledge and focusing on innovation.

In-service K1-12 teachers across Kosovo (see Fig. 1 for the frequency and distribution of school teachers across Kosovo) responded to each item by selecting one of five options, indicating the option that most accurately reflects their level of agreement or alignment with the statement. This structured approach enables a clear and measurable assessment of the self-perceived competency of each teacher in relation to the defined standards of the DigCompEdu framework.

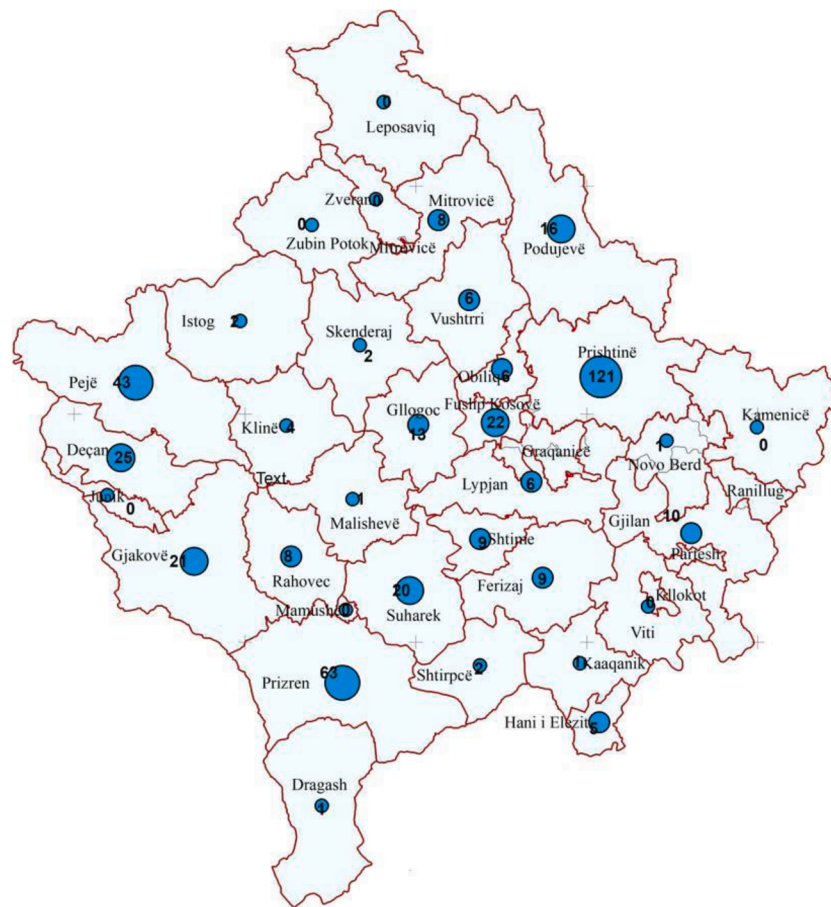


Fig. 1. Participant distribution across 26 municipalities in Kosovo.

### 3.2. Data analysis

The analysis used a combination of statistical methods to ensure accurate and meaningful insights from the data. First, reliability testing was performed using Cronbach's alpha to assess the internal consistency of the measurement instrument in six areas. This method is critical to confirm that the items within each area are measuring the same construct reliably, providing confidence in the robustness of the data. The analysis continued with normality tests; specifically, the Kolmogorov-Smirnov (KS) and Shapiro-Wilk (SW) tests were applied to determine whether the data followed a normal distribution. These tests were essential to guide the selection of appropriate statistical methods for further analysis. As a result, the analysis proceeded with non-parametric methods due to the significant deviation from normality.

The Kruskal-Wallis test was chosen as the primary method to compare differences between groups, as it does not rely on assumptions of normal distribution. This test was used to examine the variations in three factors: municipalities in Kosovo, age, and years of experience using digital technology in teaching for the six areas. Because the SELFIE for Teachers instrument employs Likert-type response scales, tied ranks were expected within the dataset; therefore, standard tie-correction procedures implemented in the statistical software were applied during the Kruskal-Wallis analyses. By using this test, the analysis could identify significant differences in responses without being constrained by the non-normal distribution of the data. For all statistically significant Kruskal-Wallis results, Dunn-Bonferroni post hoc tests were subsequently conducted to identify the specific group differences underlying the omnibus effects. The findings indicated that significant differences were primarily associated with teachers' technology-related experience across all six DigCompEdu areas, whereas age-related

differences were observed only in Areas 2 and 5. No statistically significant differences were identified across municipalities.

## 4. Results

### 4.1. Demographics

Table 4 provides a detailed breakdown of the demographic characteristics of the participants. A total of 441 participants from 26 municipalities in Kosovo participated in the assessment.

Among the 441 participants, 74% identified as females and 26% as males. The participants represented four different pre-university education levels in Kosovo: (i) K1–5: 133 participants (30%), (ii) K6–9: 199 participants (45%), (iii) General K10–12: 54 participants (12%), and (iv) Professional high schools K10–12: 55 participants (13%). The age distribution of participants reveals that 18% are within the 25 to 29 age range, 61% between the ages of 30 and 49, and followed by 21% who are over 50 years old.

The teaching experience of K1-12 teachers varies, with the largest group having 10–19 years of experience (34%). Continuing with early-career teachers with 1–5 years of experience (28%), while 23% have been teaching for over 20 years. Teachers with 6–9 years of experience comprise 14% of the group, and only 2% decided not to disclose their years of experience.

When it comes to the use of digital technology in teaching, the data indicated that the majority of K1-12 teachers are relatively new to incorporating technology into their practice. Most K1-12 teachers (43%) had 1–5 years of experience using digital tools, followed by 26% of K1-12 teachers with 10–19 years of experience. Only a small proportion had 6–9 years (17%) or more than 20 years (4%) of digital technology

**Table 4**  
Demographic information of the participants.

|                                     |                         | n   | %  |                                   |     | n  | % |
|-------------------------------------|-------------------------|-----|----|-----------------------------------|-----|----|---|
| <i>Gender</i>                       |                         |     |    | <i>School Level</i>               |     |    |   |
|                                     | Female                  | 327 | 74 | K1–5                              | 133 | 30 |   |
|                                     | Male                    | 113 | 26 | K6–9                              | 199 | 45 |   |
|                                     | Not Declared            | 1   | 0  | K10–12                            | 54  | 12 |   |
| <i>Age</i>                          |                         |     |    | <i>School</i>                     |     |    |   |
|                                     | Under 25                | 23  | 5  | Public                            | 425 | 96 |   |
|                                     | 25–29                   | 56  | 13 | Private                           | 16  | 4  |   |
|                                     | 30–39                   | 151 | 34 |                                   |     |    |   |
|                                     | 40–49                   | 119 | 27 |                                   |     |    |   |
|                                     | 50–59                   | 85  | 19 |                                   |     |    |   |
|                                     | over 60                 | 7   | 2  |                                   |     |    |   |
| <i>Years of teaching experience</i> |                         |     |    | <i>Dig.tech. use for teaching</i> |     |    |   |
|                                     | 1–5 years               | 122 | 28 | 1–5 years                         | 188 | 43 |   |
|                                     | 6–9 years               | 63  | 14 | 6–9 years                         | 75  | 17 |   |
|                                     | 10–19 years             | 148 | 34 | 10–19 years                       | 114 | 26 |   |
|                                     | 20 years and above      | 101 | 23 | 20 years and above                | 17  | 4  |   |
|                                     | I don't want to declare | 7   | 2  | I don't want to declare           | 20  | 5  |   |
|                                     |                         |     |    |                                   |     |    |   |
|                                     |                         |     |    |                                   |     |    |   |

experience. Similarly to teaching experiences, a small group of K1-12 teachers chose not to disclose their digital experience (5%).

#### 4.2. Digital competencies for K1-12 teachers

Overall, the data drew a picture of a predominantly female mid-career teaching workforce working mainly in public schools. Although many had extensive classroom experience, their use of digital technology in teaching is generally recent, highlighting the need for further professional development in this area.

To further examine the measurement properties of the DigCompEdu areas, the suitability of the data for factor analysis was assessed. The Kaiser–Meyer–Olkin (KMO) value of 0.894 and a statistically significant Bartlett's Test of Sphericity ( $\chi^2(15) = 1535.49, p < .001$ ) indicated that the data were appropriate for factor analysis. Principal Component Analysis revealed a single-factor solution, with an eigenvalue of 4.023, explaining 67.05% of the total variance. All six areas demonstrated strong factor loadings (0.730–0.876) and acceptable communalities (0.533–0.768), suggesting that each variable contributes meaningfully to the underlying construct. These results support a unidimensional structure, indicating that the DigCompEdu areas collectively reflect a common latent dimension of teachers' digital competence.

Table 5 shows the reliability analysis results, which shows that the reliability statistics of all six areas of SELFIE are between .870 and .897, indicating a strong internal consistency. In addition, a further estimate of reliability was computed as McDonald's Omega was calculated for the overall scale as  $\omega = .928$ , providing additional validation of the internal consistency of the instrument.

As shown in Table 6, the mean result for the six areas ranged from 2.823 (Area 3) to 3.325 (Area 2). Area 2 Digital Resources represented

**Table 5**  
Reliability test using Cronbach's Alpha.

| Area   | Cronbach's Alpha |
|--------|------------------|
| Area 1 | .897             |
| Area 2 | .890             |
| Area 3 | .870             |
| Area 4 | .890             |
| Area 5 | .881             |
| Area 6 | .877             |

**Table 6**  
Descriptive statistics of six DigCompEdu areas.

| DigCompEdu                                | Mean   | Std. Deviation | N   |
|-------------------------------------------|--------|----------------|-----|
| Professional Engagement                   | 2.9501 | .76225         | 441 |
| Digital Resources                         | 3.3250 | .85852         | 441 |
| Teaching and Learning                     | 2.8226 | .92294         | 441 |
| Assessment                                | 2.9947 | .78510         | 441 |
| Empowering Learners                       | 3.0121 | .98824         | 441 |
| Facilitating Learners' Digital Competence | 2.9596 | .91020         | 441 |

the highest level of perceived competence ( $M = 3.325$ ) whereas Area 3 Teaching and Learning, the lowest level ( $M = 2.823$ ). The standard deviation values indicated variability across the areas, ranging from the lowest variability with 0.762 in Area 1 Professional Engagement, to the highest variability with 0.988 in Area 5 Empowering Learners. These results highlighted Area 2 Digital Resources as the area of highest competence, whereas Area 3 Teaching and Learning emerged as the area with the lowest competence.

In addition, the data were subjected to the Kolmogorov-Smirnov (KS) and Shapiro-Wilk (SW) normality tests (see Table 7).

Based on Table 7's results, it is shown that the p-value for all six areas is  $< .001$ , indicating a significant deviation from normality for every area. Similarly, the p-values for all six areas are  $< 0.01$ , reinforcing the conclusion that the data for each area significantly deviates from a normal distribution. Area 6 Facilitating Learners' Digital Competence showed a slightly higher p-value in the Kolmogorov-Smirnov test, but it was still below the significance threshold. Thus, it also failed the normality test. These facts indicated the next step was to proceed with a non-parametric test, specifically with Kruskal-Wallis.

Table 8 presents the results of the Kruskal-Wallis test across three factors: (i) age, (ii) years of experience using digital technology in teaching, and (iii) municipalities in Kosovo for each of the six DigCompEdu areas. The reported values include the Kruskal-Wallis statistic ( $H$ ), exact p-values, and effect sizes ( $\epsilon^2$ ). The results indicate that experience with digital technology is the most consistent factor influencing teachers' digital competence, showing statistically significant differences across all six DigCompEdu areas with small to moderate effect sizes.

Regarding the municipalities factor in Kosovo, the results revealed no significant differences between K1-12 teachers in any of the areas, indicating that this factor did not have a substantial effect on these areas (see Fig. 2).

With respect to the age factor, the results indicate that statistically significant differences across age groups (Under 25, 25–39, 40–59, and 60+) were observed only in Area 2 ( $H = 30.70, p < .001$ ) and Area 5 ( $H = 18.62, p = .002$ ). In contrast, no significant differences were found for Area 1 ( $p = .171$ ), Area 3 ( $p = .080$ ), Area 4 ( $p = .354$ ), and Area 6 ( $p = .113$ ).

Inspection of the mean ranks suggests that differences across age groups are not consistent across all areas. For Area 2, younger and early-career groups as well as mid-career teachers (Under 25 and 25–39) tend

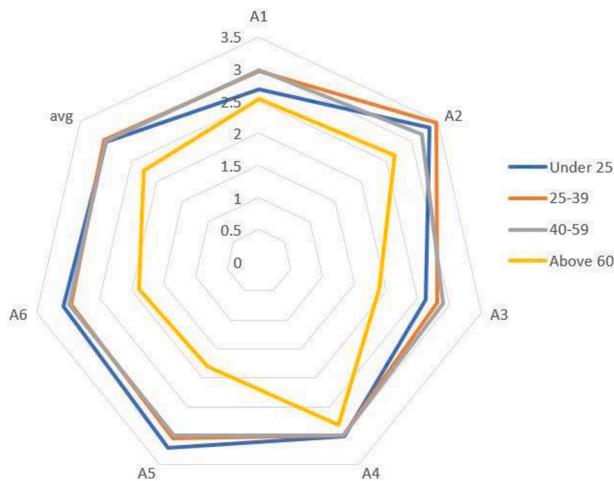
**Table 7**  
Tests of normality.

| DigCompEdu                | Kolmogorov-Smirnov |     |       | Shapiro-Wilk |     |       |
|---------------------------|--------------------|-----|-------|--------------|-----|-------|
|                           | Statistic          | df  | Sig.  | Statistic    | df  | Sig.  |
| Professional Engagement   | .089               | 441 | <.001 | .986         | 441 | <.001 |
| Digital Resources         | .101               | 441 | <.001 | .976         | 441 | <.001 |
| Teaching and Learning     | .081               | 441 | <.001 | .978         | 441 | <.001 |
| Assessment                | .130               | 441 | <.001 | .978         | 441 | <.001 |
| Empowering Learners       | .079               | 441 | <.001 | .976         | 441 | <.001 |
| Facilitating Learners' DC | .057               | 441 | .002  | .986         | 441 | <.001 |

**Table 8**

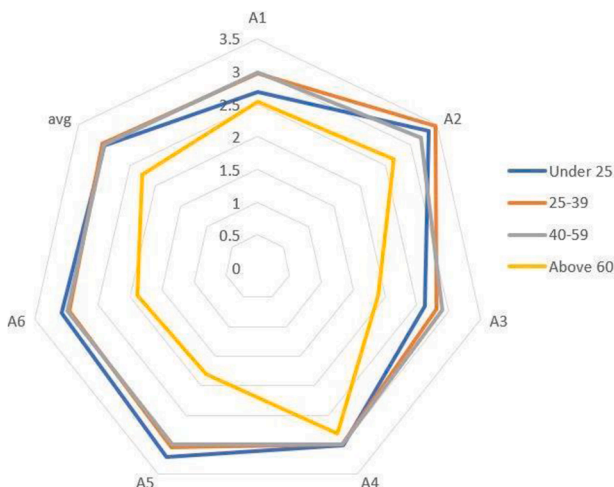
Summary of Kruskal–Wallis results across DigCompEdu areas.

| Area   | Age<br>(k = 6)<br>H, df, p, $\epsilon^2$ | Experience<br>(k = 8)<br>H, df, p, $\epsilon^2$ | Municipality<br>(k = 26)<br>H, df, p, $\epsilon^2$ |
|--------|------------------------------------------|-------------------------------------------------|----------------------------------------------------|
| Area 1 | 7.74, 5, .171, .006                      | 60.62, 7, <.001, .124                           | 18.05, 25, .840, ~0                                |
| Area 2 | 30.70, 5, <.001, .059                    | 18.06, 7, .012, .026                            | 17.83, 25, .850, ~0                                |
| Area 3 | 9.84, 5, .080, .011                      | 62.80, 7, <.001, .129                           | 26.95, 25, .359, .005                              |
| Area 4 | 5.54, 5, .354, .001                      | 41.90, 7, <.001, .081                           | 22.94, 25, .581, ~0                                |
| Area 5 | 18.62, 5, .002, .031                     | 43.75, 7, <.001, .085                           | 23.21, 25, .565, ~0                                |
| Area 6 | 8.90, 5, .113, .009                      | 32.92, 7, <.001, 0.60                           | 17.95, 25, .845, ~0                                |

**Fig. 2.** Digital competences across main municipalities in Kosovo.

to show higher mean ranks, while older groups show lower ranks, indicating lower reported competence. A similar pattern is observed in Area 5, where younger age groups (Under 25) demonstrate relatively higher competence compared to older groups, above 60 (see Fig. 3).

Fig. 4 shows digital competencies across experience with technology categories. For the years of experience using digital technology in teaching, the results presented in Table 8 indicate statistically significant differences across all DigCompEdu areas: Area 1 ( $H = 60.62, p < .001$ ), Area 2 ( $H = 18.06, p = .012$ ), Area 3 ( $H = 62.80, p < .001$ ), Area 4 ( $H = 41.90, p < .001$ ), Area 5 ( $H = 43.75, p < .001$ ), and Area 6 ( $H = 32.92, p < .001$ ). Inspection of the mean ranks reveals a consistent pattern across all areas, whereby higher levels of experience with digital technology are associated with higher mean ranks, indicating greater reported

**Fig. 3.** Digital competencies across age categories.

digital competence. When these results are considered in cumulative categories (1–3 years, 4–9 years, and above 10 years of experience), a clear gradient emerges: teachers with 1–3 years of experience show the lowest levels of digital competence, those with 4–9 years demonstrate moderate levels, and those with more than 10 years exhibit the highest levels of competence. This pattern suggests a progressive increase in digital competence with accumulated experience using digital technology in teaching. Overall, these findings indicate that experience with digital technology is a strong and consistent factor associated with differences in digital competence across all DigCompEdu areas.

The descriptive statistics of the age and technology-experience groups are presented in Tables 9 and 10 with median and interquartile range values.

These tables supplement the inference results shown in Table 8 and support the interpretation of Figs. 2, 3, and 4.

In line with the results of the Kruskal–Wallis test, the highest variation in median and IQR values was found across technology-experience groups, suggesting more differences in digital competence among teachers with different levels of experience using digital technologies. Age-related differences were less evident and were statistically significant only in Area 2 and Area 5, while no significant age-related differences were observed in the remaining DigCompEdu areas.

#### 4.3. Exploring DigCompEdu results through the lens of TPACK

The use of the TPACK framework to interpret the results provided a detailed understanding of how teachers integrated Technological Knowledge (TK), Pedagogical Knowledge (PK), and Content Knowledge (CK) across various domains of digital competence.

Table 11 aligns TPACK core components with DigCompEdu frameworks. Technological Knowledge (TK) aligned with Areas 2 Digital Resources and 6 Facilitating Learners' Digital Competence in DigCompEdu, with a mean score of 3.14. This suggested a relatively strong connection between TPACK's TK and the competency areas in DigCompEdu that emphasize the use of digital tools and the empowerment of learners.

Pedagogical Knowledge (PK) aligned with Areas 3 Teaching and Learning and 5 Empowering Learners in DigCompEdu, scoring a mean of 2.92. This indicated a moderate connection between general pedagogical practices and these DigCompEdu areas.

Content Knowledge (CK) linked to Areas 2 Digital Resources and 4 Assessment with a mean score of 3.16. This reflected a stronger relationship between CK and the use of digital tools for content delivery and assessment. Technological Pedagogical Knowledge (TPK) aligned with Areas 3 (Teaching and Learning and 5 Empowering Learners, scoring 2.92, suggesting a similar level of integration as seen with PK and indicating room for improvement in the intersection between technology and pedagogy within DigCompEdu. Technological Content Knowledge (TCK) also linked to Areas 2 Digital Resources and 4 Assessment, with a mean of 3.16, paralleling CK. This alignment showed how well TCK maps to the use of technology for subject-specific tasks and evaluation.

Pedagogical Content Knowledge (PCK) linked to Areas 3 Teaching and Learning and 5 Empowering Learners, scoring 2.92. Similarly to the results in PK and TPK, the moderate score indicated opportunities to improve how the pedagogy integrates with the content in DigCompEdu.

Technological Pedagogical Content Knowledge (TPACK) linked to Areas 3 Teaching and Learning, 4 Assessment, and 5 Empowering Learners, with a mean score of 2.94. This general integration of technology, pedagogy and content knowledge in DigCompEdu is moderate but could benefit from stronger statistical connections.

The scores for PK, PCK and TPK (all 2.92) revealed that DigCompEdu did not strongly align with the pedagogical focus of TPACK. This indicated a lack of emphasis on how digital tools and methods can transform teaching strategies and enhance learning beyond technical competence. Higher alignment scores for TK (3.14), CK (3.16) and TCK (3.16)



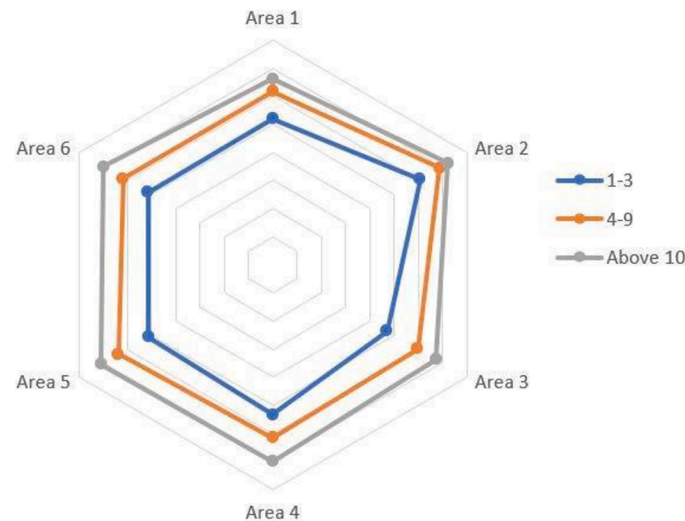


Fig. 4. Digital competencies across experience with technology categories.

**Table 9**  
Median and interquartile range (IQR) by age group.

| Area   | Under 25       | 25–29          | 30–39          | 40–49          | 50–59          | 60+            |
|--------|----------------|----------------|----------------|----------------|----------------|----------------|
| Area 1 | 2.50<br>(1.00) | 2.88<br>(1.00) | 3.00<br>(1.00) | 3.00<br>(1.00) | 3.00<br>(0.75) | 2.50<br>(0.50) |
| Area 2 | 3.33<br>(0.83) | 4.00<br>(1.08) | 3.33<br>(1.67) | 3.33<br>(1.33) | 3.00<br>(1.33) | 2.33<br>(0.67) |
| Area 3 | 2.50<br>(1.25) | 2.75<br>(1.31) | 2.75<br>(1.50) | 3.00<br>(1.50) | 2.75<br>(1.00) | 1.75<br>(1.25) |
| Area 4 | 3.00<br>(0.50) | 3.00<br>(1.00) | 3.00<br>(1.17) | 3.00<br>(1.17) | 3.00<br>(1.00) | 3.00<br>(0.33) |
| Area 5 | 3.33<br>(0.67) | 3.00<br>(1.75) | 3.00<br>(1.50) | 3.00<br>(1.67) | 2.67<br>(1.00) | 1.67<br>(0.67) |
| Area 6 | 3.00<br>(1.20) | 2.80<br>(1.25) | 3.00<br>(1.30) | 3.00<br>(1.00) | 3.00<br>(1.20) | 1.60<br>(1.10) |

indicated that DigCompEdu prioritized the knowledge of digital tools and their application in subject-specific contexts. However, this tool- and content-centric approach could lead K1-12 teachers to focus on the use of technology without fully considering its pedagogical implications, limiting meaningful teaching innovation and creativity.

The results did not indicate how effectively DigCompEdu integrated emerging technologies, such as artificial intelligence, virtual reality, or gamification, into educational frameworks. This limitation suggested the need for an initiative to enhance the DigCompEdu framework by integrating emerging technologies, as its current structure may otherwise hinder the preparation of K1-12 teachers for future innovations.

## 5. Discussion

This section explores the findings of the study and addresses each research question. By analyzing the digital competencies of K1-12 teachers in Kosovo, this discussion offers insights into their current

proficiency levels, the factors that influence their digital skills, and the alignment of DigCompEdu with complementary frameworks like TPACK.

### 5.1. RQ1: what are the ICT competence level of Kosovo's K1-12 teachers based on DigCompEdu framework ?

The results indicate that Kosovo's K1-12 teachers perceive their highest competence in Digital Resources (Area 2) and their weakest competence in Teaching and Learning (Area 3). This aligns with prior findings by Trindade and Moreira (2020) on Portuguese teachers, where pedagogical competences, including assessment and empowering learners, were similarly identified as weaker areas. The strong performance in Area 2 suggests that teachers in Kosovo are relatively confident in selecting, managing, and creating digital resources. However, the lower scores in Area 3 highlight the challenges in integrating these resources into teaching practices effectively. This is also confirmed by the low scores on pedagogical knowledge (PK).

These results reflect a tool-centric approach emphasized by DigCompEdu, as noted in the literature. While the framework successfully evaluates technical skills, it places less emphasis on pedagogical

**Table 11**  
The alignment of TPACK with DigCompEdu framework.

| TPACK | DigCompEdu      | Mean |
|-------|-----------------|------|
| TK    | Area 2 and 6    | 3.14 |
| PK    | Area 3 and 5    | 2.92 |
| CK    | Area 2 and 4    | 3.16 |
| TPK   | Area 3 and 5    | 2.92 |
| TCK   | Area 2 and 4    | 3.16 |
| PCK   | Area 3 and 5    | 2.92 |
| TPACK | Area 3, 4 and 5 | 2.94 |

**Table 10**  
Median and interquartile range (IQR) by technology experience.

| Area   | No experience | 1–3 years   | 4–5 years   | 6–9 years   | 10–14 years | 15–19 years | 20+ years   | Prefer not to declare |
|--------|---------------|-------------|-------------|-------------|-------------|-------------|-------------|-----------------------|
| Area 1 | 2.00 (0.62)   | 2.75 (1.00) | 2.75 (0.81) | 3.00 (1.25) | 3.25 (0.75) | 3.25 (1.25) | 3.50 (1.00) | 2.62 (0.81)           |
| Area 2 | 2.67 (1.00)   | 3.33 (1.33) | 3.33 (1.33) | 3.33 (1.33) | 3.67 (1.00) | 3.33 (1.00) | 3.67 (1.33) | 3.00 (0.75)           |
| Area 3 | 2.00 (1.25)   | 2.50 (1.25) | 2.75 (1.25) | 3.00 (1.50) | 3.25 (1.25) | 3.25 (1.00) | 3.00 (1.00) | 1.88 (1.25)           |
| Area 4 | 2.67 (0.67)   | 3.00 (1.00) | 3.00 (1.33) | 3.00 (1.00) | 3.00 (1.00) | 3.33 (1.33) | 3.00 (1.00) | 2.33 (1.00)           |
| Area 5 | 2.33 (1.50)   | 2.67 (1.67) | 3.00 (1.33) | 3.00 (1.50) | 3.33 (1.33) | 3.33 (1.33) | 3.00 (1.33) | 2.50 (1.33)           |
| Area 6 | 2.40 (1.50)   | 2.80 (1.00) | 2.80 (1.20) | 3.00 (1.30) | 3.20 (1.00) | 3.20 (1.20) | 3.00 (1.00) | 2.20 (1.55)           |

integration, which is crucial for meaningful use of digital tools in classrooms. This gap highlights the need for training programs that focus not only on the use of technology but also on its pedagogical applications. Research has established that in order to make affective and integrated use of digital technology in the classroom, basic and advanced digital competencies are necessary; that is, the translation of technical knowledge into practical educational intervention as emphasized in [49].

*5.2. RQ2: how do digital competencies, as defined by the DigCompEdu framework, vary between different age categories, levels of technology experience, and major cities in Kosovo?*

The results reveal that younger teachers (those under 25 and the 25–39 age group) demonstrated higher reported competence in Digital Resources (Area 2) compared to their older colleagues. This pattern suggests that younger educators, often categorized as digital natives, may possess a higher level of comfort and fluency in selecting, managing, and creating digital content due to early and sustained exposure to these technologies. Conversely, teachers in the older age categories (above 60) scored significantly lower in this area. This indicates that while classroom experience is vital for general pedagogy, it does not automatically translate to high proficiency in the technical management of digital resources. However, a high level of technical comfort does not immediately result in advanced pedagogical integration. This is consistent with the developmental journey from pre-service to experienced in-service teacher, which typically moves from teacher-centered to student-centered learning [50]. While younger teachers may excel in the 'what' (CK) and 'how' (TK) of digital tools, they may still be in a development phase regarding the complex integration of technology for student-centered learning (TPACK), a skill that often matures with sustained classroom practice.

The analysis revealed differences in digital competence across age groups and levels of experience with digital technology; however, these findings should be interpreted with caution. Although some differences reached statistical significance, the magnitude of these differences appears modest, suggesting limited practical impact on teachers' overall competence profiles. In particular, the younger and mid-career teachers (25–39) reported the highest competence in selected areas such as Digital Resources and Empowering Learners. While this pattern may reflect higher fluency among younger age groups, the observed differences are not substantial enough to indicate clear categorical distinctions between age groups, as the magnitude of these differences appears modest. Similarly, although experience with digital technology in teaching was associated with differences across all competence areas, these differences likely reflect gradual development rather than discrete competence levels. These findings suggest that digital competence is shaped by continuous professional experience rather than sharply differentiated by demographic categories. Therefore, interpretations should avoid overemphasizing group differences and instead focus on the broader pattern that sustained engagement with digital technologies contributes incrementally to competence development.

Experience with digital technology in teaching also significantly influenced all competency areas. Teachers with 10–19 years of experience using technology performed better, emphasizing the importance of sustained exposure to digital tools. These findings validate prior studies by [30] highlighting that experience with technology is a critical factor in developing digital competence. The lack of differences across municipalities in Kosovo suggests that digital competence is not region-specific but influenced more by individual technological experience, age-related digital fluency, and professional development opportunities.

*5.3. RQ3: how does the DigCompEdu framework compare to TPACK in integrating emerging technologies to prepare K1-12 teachers for future innovations?*

The alignment of DigCompEdu with TPACK revealed complementary strengths and weaknesses. While DigCompEdu emphasizes technical and individual skills, TPACK offers a more holistic approach by integrating Technological Knowledge (TK), Pedagogical Knowledge (PK), and Content Knowledge (CK) into a unified framework. The stronger alignment of DigCompEdu with TK (mean score: 3.14) and CK (mean score: 3.16) reflects its focus on tool-centric and content-specific competencies. However, a weaker alignment with PK and TPK (both with mean scores of 2.92) highlights the need for a stronger pedagogical emphasis, as we have mentioned above when answering RQ1. The translation of how to use technology into actual educational, effective practice by the teacher is not trivial. Therefore, the Kosovar government and universities should join forces in establishing a digital skills educator program for pre- and in-service teachers with a strong emphasis on the intersection of TPACK knowledge.

To further contextualize these findings, TPACK is used here as an interpretive framework rather than an empirically measured construct. The observed pattern, which is relatively higher competence in Digital Resources and comparatively lower competence in teaching and learning, can be interpreted as reflecting stronger Technological Knowledge (TK) and Technological Content Knowledge (TCK), alongside comparatively weaker Technological Pedagogical Knowledge (TPK). In other words, teachers appear more confident in accessing and managing digital tools than in integrating these tools into pedagogically meaningful learning processes. This interpretation is consistent with prior research suggesting that teachers' digital development often begins with tool-oriented competence before progressing toward deeper pedagogical integration. However, it is important to emphasize that this alignment does not constitute direct evidence of teachers' TPACK levels but rather provides a theoretically informed lens for understanding the pedagogical implications of the DigCompEdu-based results.

This tool-centric approach, while effective in improving technical skills, may limit the transformative potential of digital tools in education. As noted in the literature, frameworks like TPACK provide a more integrated perspective, ensuring that digital tools are not only used effectively but also innovatively to enhance pedagogy and learning outcomes. The results suggest that DigCompEdu could benefit from incorporating elements of TPACK to better address emerging technologies such as artificial intelligence, virtual reality, and gamification.

It is important to distinguish between findings directly supported by the data and broader interpretations. The empirical results of this study are based on self-reported measures of digital competence and non-parametric group comparisons. Therefore, conclusions regarding pedagogical practices, classroom effectiveness, or the integration of emerging technologies should be interpreted cautiously. While such interpretations are informed by theory, they extend beyond what can be directly inferred from the data.

*5.4. Implications and recommendations*

This study's results provide significant insights into the digital competence levels of K1-12 teachers in Kosovo, evaluated using the DigCompEdu methodology. Although teachers exhibit considerable proficiency in managing and developing digital resources (Area 2), their effectiveness in incorporating these technologies into pedagogical practices (Area 3) is somewhat less robust. Moreover, this study also identified a gap in relation to the significant shortcoming of the DigCompEdu paradigm, which prioritizes technical abilities while inadequately addressing the pedagogical integration essential for successful classroom use.

A notable finding is also the influence of experience on the development of digital competence. Teachers with more technological

experience, namely those with 10 to 19 years of experience, surpassed their less experienced peers in all skill domains. This indicates that prolonged exposure to digital technologies enhances confidence and proficiency in their use. Moreover, the results indicate that DigCompEdu may enhance its framework by integrating aspects of TPACK, especially when new technologies like artificial intelligence, virtual reality, and gamification gain prominence in education.

Given the mentioned implications, several recommendations emerge to address the identified gaps and challenges.

- There is a distinct need to redirect the emphasis of teacher training programs towards pedagogical integration. Technical proficiency is essential; nevertheless, it must be augmented by tactics that use digital resources to promote collaborative learning, self-regulated learning, and active student participation. Training programs must emphasize Areas 3 (Teaching and Learning) and 5 (Empowering Learners), providing teachers with effective strategies to establish technology-enhanced, student-centered learning environments. To achieve this, Kosovo's education system might gain from implementing hybrid frameworks that integrate the advantages of DigCompEdu and TPACK. Integrating TPACK's comprehensive framework into teacher education programs enables both pre-service and in-service teachers to cultivate a profound comprehension of aligning technological applications with educational objectives and material dissemination. This would also equip them to integrate new technology into their pedagogical methods more efficiently.
- Professional development programs must be customized according to instructors' technological proficiency and determined needs and realities in classroom settings. The offering of these programs should not be planned in isolation but rather within a sustainable offer of in-service programming within a national system of in-service teacher education. This is important to view these as continuous and long-term interventions rather than ad hoc inputs that are short-term and not able to follow teachers' involvement into absorbing these concepts. The important part of the system is to integrate reflective practices for teachers to engage in understanding their practice and engaging in action-oriented practices. When these concepts are integrated with a sustainable mechanism of in-service education, chances are that better results are achieved in classroom practices. Novice teachers or those with no classroom experience might benefit from fundamental training in handling digital resources and basic integration approaches. Advanced programs concentrating on new pedagogical uses of technology might assist experienced instructors in evolving from skilled users to leaders in digital education.
- Continuous professional development (CPD) programs must be established to provide sustained assistance to teachers throughout their careers. These programs may include mentoring opportunities whereby senior teachers assist less experienced colleagues in incorporating technology into their pedagogical approaches. Moreover, peer-to-peer learning networks may facilitate cooperation among teachers, allowing them to exchange best practices and effective ways for using digital resources.
- At the policy level, it is essential for the Kosovar government and universities to emphasize pedagogical training in conjunction with technological skill development within teacher education curriculum. National policies have to provide financing for professional development initiatives that prioritize the significant incorporation of technology into pedagogical methods. Moreover, educational institutions have to be furnished with advanced technology, like AI-driven tools and virtual reality kits, to enable teachers to engage in practical experimentation with cutting-edge innovations in the classrooms.
- Ultimately, systems for assessing and evaluating the efficacy of training programs should be implemented. Self-assessment instruments such as SELFIE for Teachers or longitudinal studies may facilitate the monitoring of advancements over time, especially in

Areas 3 (Teaching and Learning) and 5 (Empowering Learners). This would guarantee that training activities are both effective and perpetually enhanced in response to instructor feedback and changing educational requirements.

In summary, the findings of this study underscore that for digital transformation to be effective in Kosovo, policy and professional development must shift from a focus on technical tool acquisition to deep pedagogical integration. The data indicates that while teachers are proficient in managing digital resources, they struggle to translate these skills into student-centered learning environments. Therefore, it is recommended that Kosovo adopt hybrid training frameworks that bridge the technical competencies of DigCompEdu with the pedagogical depth of the TPACK model. Furthermore, professional development should move away from ad-hoc workshops toward sustainable, long-term programs that incorporate mentoring and peer-to-peer networks to support gradual competence development. At the policy level, this requires targeted financing for both infrastructure, such as AI and virtual reality, and the implementation of continuous evaluation systems to ensure that teacher training remains responsive to the evolving demands of the digital landscape. By focusing on these pillars, Kosovo can transition digital technology from a mere classroom tool into a transformational instrument for educational innovation.

### 5.5. Limitations

Despite the insights provided, this study is subject to several methodological limitations that warrant clearer emphasis.

First, the reliance on the SELFIE for Teachers tool means the data is entirely self-reported, introducing the inherent risk of self-perception bias and social desirability; teachers may subconsciously overestimate their competencies to align with professional expectations. Research has in fact shown that high SELFIE for Teachers scores correspond with findings that self-assessments tend to represent perceived confidence, professional orientation, and intended practice, rather than serving solely as indicators of enacted classroom behaviour [51,52], which were not measured in our study.

Second, the use of snowball sampling to reach 441 participants across 26 municipalities was necessary given the lack of existing teacher registries for Kosovo, but this non-probability method limits the representativeness of the sample and the generalizability of the findings to the entire Kosovar teaching workforce.

Third, the linkage between DigCompEdu and TPACK in this study should be viewed as an interpretive conceptual limitation; while we used TPACK as a lens to understand the findings, we did not directly empirically measure the seven TPACK components, which may limit the depth of our pedagogical insights. Additionally, digital competence requirements and manifestations likely vary across educational contexts, as suggested by researchers' work on context-sensitive competence frameworks [53], representing an important direction for future research.

Finally, the cross-sectional design of this study provides only a snapshot of teacher competence at a single point in time. This design prevents us from making causal inferences regarding the drivers of digital competence or providing a developmental interpretation of how these skills evolve over a career.

## 6. Conclusion and future work

Since digital competencies are essential in a modern society, the primary goal is to equip teachers with advanced digital skills to enhance education quality and stay aligned with global developments. As a critical part of a teacher's professional toolkit, digital competence empowers them to navigate the complexities of technology-driven education and prepare students for future challenges.

The assessment of K1-12 teachers' digital competencies in Kosovo

reveals a mixed landscape, with strengths in resource utilization but notable gaps in pedagogical application. The results emphasize the importance of targeted professional development programs that address these gaps, especially in the "Teaching and Learning" domain, to enable teachers to effectively integrate technology into classroom practices. Aligning the DigCompEdu and TPACK frameworks offers a holistic approach, ensuring both skill acquisition and contextual application. However, current frameworks must evolve to incorporate emerging technologies, such as artificial intelligence and virtual reality, to future-proof education. This study highlights the urgency of developing context-sensitive digital competency frameworks and tailored training initiatives in Kosovo that cater to diverse educator needs. These efforts are critical for fostering innovation and achieving equitable digital transformation across Kosovo's education system.

The findings of this study should be interpreted within the limits of self-reported data and cross-sectional design. While the results provide valuable insight into teachers' perceived digital competence, they do not establish causal relationships or directly reflect classroom practice. The observed patterns indicate that teachers demonstrate relatively stronger competence in using digital resources, while pedagogical integration remains an area for further development. However, differences across demographic groups, although statistically significant in some cases, appear modest and should not be over-interpreted. The use of TPACK as an interpretive lens provides additional theoretical insight into these patterns, suggesting a need to strengthen the integration of technology with pedagogy and content. At the same time, this interpretation remains conceptual and would benefit from future studies that directly measure TPACK constructs. Finally, while the study highlights the importance of preparing teachers for emerging technologies, such implications extend beyond the present data and should be explored in future research through empirical investigation.

While this study provides a foundational understanding of K1-12 teachers' digital competencies in Kosovo, several areas require further exploration for future work:

- **Longitudinal Studies** – Future research should conduct longitudinal assessments to track teachers' digital competence development over time, evaluating the effectiveness of interventions aimed at improving pedagogical integration of digital tools.
- **Classroom Observations and Student Outcomes** — This study relied on self-assessment data, which, while valuable, is subject to bias. Future research should include direct classroom observations and student performance data to assess how digital competency translates into teaching effectiveness and student learning outcomes.
- **Emerging Technologies** – The DigCompEdu framework currently does not fully address the integration of artificial intelligence (AI), virtual reality (VR), or gamification in education. Future research should explore how these emerging technologies can be incorporated into teacher training and their impact on teaching and learning.
- **Cross-Cultural Comparisons** – Given that Kosovo is still developing its digital education strategy, comparative studies with Nordic countries (e.g., Finland, Sweden, Norway) could provide valuable insights into best practices for professional development and policy adaptation.

We believe that the further exploration of these gaps could provide a more comprehensive understanding of how digital competence develops among K-12 teachers. This knowledge would be valuable in ensuring that Kosovo's education system remains adaptive and forward-thinking as technology continues to evolve.

#### CRediT authorship contribution statement

**Krenare Pireva Nuci:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Fisnik Dalipi:** Writing – review & editing,

Writing – original draft, Methodology, Conceptualization. **Italo Masiello:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Ismail Celik:** Writing – review & editing, Writing – original draft, Methodology. **Jari Laru:** Writing – review & editing, Writing – original draft. **Blerim Saqipi:** Writing – review & editing, Writing – original draft. **Arsim Susuri:** Writing – review & editing, Data curation. **Ali Shariq Imran:** Writing – review & editing, Conceptualization.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### References

- [1] Masiello I, Fixsen DL, Nordmark S, Mohseni Z, Holmberg K, Rack J, Davidsson M, Andersson-Gidlund T, Augustsson H. Digital transformation in schools of two southern regions of Sweden through implementation-informed approach: a mixed-methods study protocol. *PLOS One* 2023;18(12):e0296000. <https://doi.org/10.1371/journal.pone.0296000>.
- [2] Van der Vlies R. Digital strategies in education across OECD countries: exploring education policies on digital technologies. *Tech Rep* 2020. <https://doi.org/10.1787/33dd4c26-en>.
- [3] S. Europaea, Key competences for lifelong learning in the European schools, Office of the secretary-general of the European schools, pedagogical development unit: Brussels, Belgium 72 (2018).
- [4] Lakkala M, Ilomäki L, Kantosalo A. Which areas of digital competence are important for a teacher. *Int J Soc Sci Educ* 2011. URL, <http://hdl.handle.net/10138/154427>.
- [5] Van Laar E, Van Deursen AJ, Van Dijk JA, De Haan J. The relation between 21st-century skills and digital skills: a systematic literature review. *Comput Hum Behav* 2017;72:577–88. <https://doi.org/10.1016/j.chb.2017.03.010>.
- [6] Pedaste M, Kallas K, Baical A. Digital competence test for learning in schools: development of items and scales. *Comput Educ* 2023;203:104830. <https://doi.org/10.1016/j.compedu.2023.104830>.
- [7] A. Bourgeois, P. Birch, O. Davydovskaia, Eurydice brief digital education at school in Europe (2019). [doi:10.2797/339457](https://doi.org/10.2797/339457).
- [8] Redecker C, Punie Y. European framework for the digital competence of educators: digCompEdu. *Tech Rep* 2017. <https://doi.org/10.2760/159770>.
- [9] Ortega-Sánchez D, Gómez-Trigueros IM, Trestini M, Pérez-González C. Self-perception and training perceptions on teacher digital competence (tdc) in Spanish and French university students. *Multimodal Technol Interact* 2020;4(4):74. <https://doi.org/10.3390/mti4040074>.
- [10] P. C. Stephanie, Digcomp into action: get inspired, make it happen. A user guide to the European digital competence framework (2018). [doi:10.2760/112945](https://doi.org/10.2760/112945).
- [11] M.J. Koehler, P. Mishra, K. Kereluik, T.S. Shin, C.R. Graham, The technological pedagogical content knowledge framework (2014) 101–111 [doi:10.1007/978-1-4614-3185-59](https://doi.org/10.1007/978-1-4614-3185-59).
- [12] Heintz F, Mannila L, Nordén L-Å, Parnes P, Regnell B. Introducing programming and digital competence in Swedish k-9 education. In: Dagienė V, Hellas A, editors. *Informatics in schools: focus on learning programming*. Cham: Springer International Publishing; 2017. p. 117–28. <https://doi.org/10.1007/978-3-319-71483-710>.
- [13] OECD. Shaping digital education: enabling factors for quality, equity and efficiency. OECD Publishing; 2023. <https://doi.org/10.1787/bac4dc9f-en>.
- [14] Çebi A, Nebioğlu N, Koçhan A. Key factors behind the differentiation of schools' digital capacities: a mixed methods investigation of stakeholders' collective reflection using the selfie tool. *Educ Inf Technol* 2026;31:2679–712. <https://doi.org/10.1007/s10639-025-13886-x>.
- [15] Horváth L, Pintér TM, Misley H, Dringó-Horváth I. Validity evidence regarding the use of digcompedu as a self-reflection tool: the case of Hungarian teacher educators. *Educ Inf Technol* 2025;30(1):1–34. <https://doi.org/10.1007/s10639-024-12914-6>.
- [16] Cachia R, Pokropek A, Giannoutsou N. Supporting the monitoring of the digital capacity of schools through optimal shortening of the selfie tool. *Comput Educ* 2024;208:104938. <https://doi.org/10.1016/j.compedu.2023.104938>.
- [17] Wang H, Li X, Zhang C, Kang Z. The constructing and empirical research on primary and secondary school teachers' digital competencies in the age of artificial intelligence. *Discov Artif Intell* 2026;6. <https://doi.org/10.1007/s44163-026-00838-8>.
- [18] Buckingham D. Defining digital literacy-what do young people need to know about digital media? *Nord J Digit Lit* 2015;10(Jubileumsnum-mer):21–35. <https://doi.org/10.18261/ISSN1891-943X-2015-Jubileumsnummer-03>.
- [19] Gilster P. Digital literacy. URL, <https://archive.org/details/digitalalliteracy0000gilster>; 1997.
- [20] Snavey L, Cooper N. The information literacy debate. *J Acad Librariansh* 1997;23(1):9–14. [https://doi.org/10.1016/S0099-1333\(97\)90066-5](https://doi.org/10.1016/S0099-1333(97)90066-5).
- [21] McMillan S. Literacy and computer literacy: definitions and comparisons. *Comput Educ* 1996;27(3–4):161–70. [https://doi.org/10.1016/S0360-1315\(96\)00026-7](https://doi.org/10.1016/S0360-1315(96)00026-7).



- [22] Harrison C. Critical internet literacy: what is it, and how should we teach it? *J Adolesc Adult Lit* 2018;61(4):461–4. <https://doi.org/10.1002/jaal.713>.
- [23] Livingstone S. What is media literacy? *Intermedia* 2004;32(3):18–20. URL, <http://eprints.lse.ac.uk/1027/>.
- [24] Heydon RM. Making meaning together: multi-modal literacy learning opportunities in an inter-generational art programme. *J Curric Stud* 2007;39(1): 35–62. <https://doi.org/10.1080/00220270500422665>.
- [25] Menon D, Sowmya B, Romero M, Viéville T. Going beyond digital literacy to develop computational thinking in k-12 education. *Epistemological approaches to digital learning in educational contexts*. Routledge; 2020. p. 17–34. <https://doi.org/10.4324/9780429319501-2>.
- [26] Falloon G. From digital literacy to digital competence: the teacher digital competency (TDC) framework. *Educ Technol Res Dev* 2020;68(5):2449–72.
- [27] Uerz D, Volman M, Kral M. Teacher educators' competences in fostering student teachers' proficiency in teaching and learning with technology: an overview of relevant research literature. *Teach Teach Educ* 2018;70:12–23. <https://doi.org/10.1016/j.tate.2017.11.005>.
- [28] Lund A, Furlberg A, Bakken J, Engelen KL. What does professional digital competence mean in teacher education? *Nord J Digit Lit* 2014;9(4):280–98. <https://doi.org/10.18261/ISSN1891-943X-2014-04-0>.
- [29] Wang W, Schmidt-Crawford D, Jin Y. Preservice teachers' tpack development: a review of literature. *J Digit Learn Teach Educ* 2018;34(4):234–58. <https://doi.org/10.1080/21532974.2018.1498039>.
- [30] Janssen J, Stoyanov S, Ferrari A, Punie Y, Pannekeet K, Sloep P. Experts' views on digital competence: commonalities and differences. *Comput Educ* 2013;68:473–81. <https://doi.org/10.1016/j.compedu.2013.06.008>.
- [31] García-Vandewalle García JM, García-Carmona M, Trujillo Torres JM, Moya Fernández P. Analysis of digital competence of educators (DigCompEdu) in teacher trainees: the context of Melilla, Spain. *Technol Knowl Learn* 2023;28(2):585–612. <https://doi.org/10.1007/s10758-021-09546-x>.
- [32] Bussesund ES, McGarr O, Engen BK. Teacher educators' discursive enactment of professional digital competencies. *J Educ Policy* 2025;40:266–87. <https://doi.org/10.1080/02680939.2024.2379821>.
- [33] Kelentrić M, Helland K, Arstorp A-T. Professional digital competence framework for teachers. *Nor Cent ICT Educ* 2017;134(1):1–74.
- [34] Siddiq F, Hatlevik OE, Olsen RV, Thronsdén I, Scherer R. Taking a future perspective by learning from the past—a systematic review of assessment instruments that aim to measure primary and secondary school students' ICT literacy. *Educ Res Rev* 2016;19:58–84. <https://doi.org/10.1016/j.edurev.2016.05.002>.
- [35] Brandhofer G, Miglbauer M. Digital competences for teachers-the Digi. Komp model in an international comparison and in the practice of Austrian teacher training. *Int J Educ* 2020;8(4):55–69. <https://doi.org/10.5121/ije.2020.8406>.
- [36] Feng L, Xue S. Using the digcompedu framework to conceptualize teachers' digital literacy. *Educ J* 2023;12(3):103–8. <https://doi.org/10.11648/j.edu.20231203.14>.
- [37] Trindade SD, Moreira JA. Assessment of high school teachers on their digital competences. *Magis: Rev Int Investig Educ* 2020;13(1):1–21.
- [38] Nordmark S, Augustsson H, Davidsson M, Andersson-Gidlund T, Holmberg K, Mohseni Z, Rack J, Masiello I. Piloting systematic implementation of educational technology in Swedish k-12 schools—two-years-in report. *Glob Implement Res Appl* 2024;4(3):309–23. <https://doi.org/10.1007/s43477-024-00130-w>.
- [39] Hamilton ER, Rosenberg JM, Akcaoglu M. The substitution augmentation modification redefinition (SAMR) model: a critical review and suggestions for its use. *TechTrends* 2016;60:433–41. <https://doi.org/10.1007/s11528-016-0091-y>.
- [40] Harris JB, Phillips M, Koehler MJ, Rosenberg JM. Editorial 33 (3) : TPCK/TPACK research and development: past, present, and future directions. *Australas J Educ Technol* 2017;33(3). <https://doi.org/10.14742/ajet.3907>.
- [41] R.R. PuenteDura, Samr: a contextualized introduction, lecture at Pine Cobble School. Retrieved March 13 (2014) 2014.
- [42] Mishra P, Koehler MJ. Technological pedagogical content knowledge: a framework for teacher knowledge. *Teach Coll Rec* 2006;108(6):1017–54. <https://doi.org/10.1111/j.1467-9620.2006.00668>.
- [43] Horváth L, Pintér TM, Misléy H, Dringó-Horváth I. Validity evidence regarding the use of DigCompEdu as a self-reflection tool: the case of Hungarian teacher educators. *Educ Inf Technol* 2025;30:1–34. <https://doi.org/10.1007/s10639-024-12914-6>.
- [44] Momdjan L, Manegre M, Gutiérrez-Colón M. Bridging the digital competence gap: a comparative study of preservice and in-service teachers in Lebanon using the DigCompEdu framework. *Technol Knowl Learn* 2025;30:655–83. <https://doi.org/10.1007/s10758-024-09794-7>. –1670.
- [45] Suzer E, Koc M. Teachers' digital competency level according to various variables: a study based on the European digcompedu framework in a large Turkish city. *Educ Inf Technol* 2024;29:22057–83. <https://doi.org/10.1007/s10639-024-12711-1>.
- [46] Momdjan L, Manegre M, Gutiérrez-Colón M. Assessing and bridging the digital competence gap: a comparative study of Lebanese student teachers and in-service teachers using the DigCompEdu framework. *Discov Educ* 2024;3(1):1–23. <https://doi.org/10.1007/s44217-024-00308-2>.
- [47] Cohen L, Manion L, Morrison K. Research methods in education. 8th ed. routledge; 2017. <https://doi.org/10.4324/9781315456539>.
- [48] Bryman A. Social research methods. Oxford university press; 2016.
- [49] E.B. Mandinach, L.M. Abrams, Data literacy and learning analytics, the handbook of learning analytics (2022) 196–204doi:10.18608/hla22.019.
- [50] Mascolo MF. Beyond student-centered and teacher-centered pedagogy: teaching and learning as guided participation. *Pedagogy Hum Sci* 2009;1(1):3–27. URL, <https://scholarworks.merrimack.edu/phs/vol1/iss1/6>.
- [51] González-Mujico FdL. Measuring student and educator digital competence beyond self-assessment: developing and validating two rubric-based frameworks. *Educ Inf Technol* 2024;29(11):13299–324.
- [52] Maderick JA, Zhang S, Hartley K, Marchand G. Preservice teachers and self-assessing digital competence. *J Educ Comput Res* 2016;54(3):326–51.
- [53] Ghomi M, Redecker C. Digital competence of educators (digcompedu): Development and evaluation of a self-assessment instrument for teachers' digital competence. *CSEDU*; 2019. p. 541–8.